

Multinomial Logistic Regression VS Perceptron: A UniMorph Classification Problem

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Abstract

This (mini) paper aims to show the difference in performance of two algorithms, a simple perceptron and a multinomial logistic regression model, evaluated over their prediction on the UniMorph (Batsuren et al. 2022) data.

1 Introduction

Morphology, at its core, is the linguistic system responsible for classification of words into categories such as nouns, verbs, adjectives, and more, via a finite set of structural patterns that the human brain learns to recognize and reapply. This study investigates whether computational models can replicate this cognitive ability: can a perceptron and a multinomial logistic regression (MLR) model classify words based on their internal structure as well as we can? Well, not really, but one might be better at learning than the other. The evaluation focuses on four languages: Italian, Swedish, Norwegian, and Northern Sami. Particular attention is given to the MLR model's learning trajectory in Norwegian, as it is the only language among the four where classification accuracy consistently improves with each training epoch, rather than declining after the initial epoch.

2 Methodology

The algorithms were implemented using pure Python, with the help of Numpy for more efficient softmax calculations in the case of the MLR model. The code for multinomial logistic regression follows the one outlined by Jurafsky and Martin (2025, Section 5.3 and 5.8) and the bootstrap test follows their guidelines for statistical significance testing (Jurafsky and Martin 2025, Section 4.9).

3 Data

The data is provided by the UniMorph (Batsuren et al. 2022) database, available at <https://unimorph.github.io/>,

where every language is analyzed word by word, and every line in the file yields the word preceded by its lemma and followed by its morphological tags, for example: "carpmes carpmesarnas N;GEN;PL;DEF". Training and testing sets for every language were randomly sampled from their files using a 80/20 split, but one must keep in mind that the amount of data was different for every language, which certainly affects the performance of the models. Italian had the biggest collection with almost 510.000 words, followed by Swedish (around 130.000 words) and Norwegian (almost 78.000), while Northern Sami came last with almost 63.000 examples.

4 Models and Evaluation

Both algorithms employ an identical set of features, namely they extract a set of prefixes and suffixes from the given word (up to 5 letters) and attempt to associate these with the appropriate tagset. The primary distinction between the two models lies in their learning and prediction mechanisms. The perceptron updates its weights using a fixed increment of 1, whereas the MLR model relies on gradient descent for optimization. In terms of prediction, the perceptron selects the highest-scoring tag by summing the weights associated with each class, while the MLR model generates a probability distribution over possible tags using a softmax function. For the MLR model, the learning rate was fixed at 0.2. The number of training epochs, however, was tuned individually for each language within a range of 1 to 10 to identify the optimal performance for both models. Model performance was assessed via bootstrap resampling, using 10,000 samples drawn with replacement from the models' prediction outputs. Each sample matched the size of the original test set. The null hypothesis (H_0) assumed the MLR model did not offer any performance advantage over the perceptron. To this

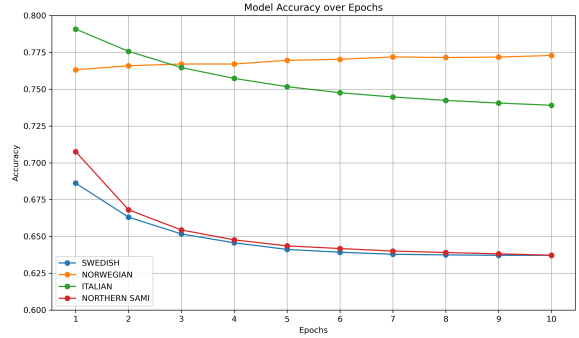
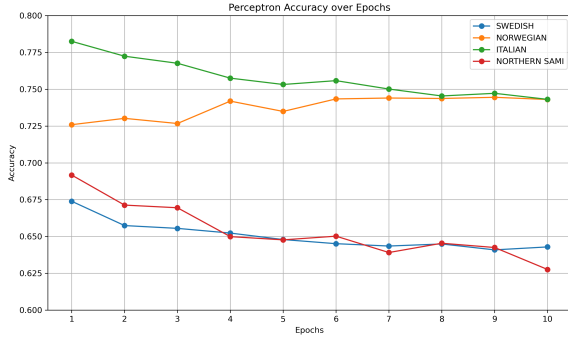


Figure 1: Accuracy of perceptron (left) and MLR model (right) for every language over 10 epochs.

end, the performance difference was calculated as $\Delta x = \text{MLR}(x) - \text{Perceptron}(x)$ for each resampled set.

5 Results

The results indicate that the MLR model outperforms the perceptron across all four languages, as evidenced by highly significant p-values (refer to Tables 1 and 2). For Italian, Swedish, and Northern Sami, both models achieved their highest performance after a single training epoch. In contrast, performance on Norwegian continued to improve with additional training, reaching its peak at 10 epochs (refer to Figure 1). Furthermore, the MLR model exhibits a smoother and more consistent learning trajectory, whereas the perceptron demonstrates a more erratic and less stable pattern of development. Nevertheless, both models generally follow similar performance trends across epochs.

6 Conclusions

Based on the test results, the MLR model appears to have superior learning capabilities compared to the perceptron. It emerges as the more suitable candidate for predicting morphological categories using UniMorph data (Batsuren et al. 2022). However, this advantage may be offset by its higher computational and time requirements. In fact, its performance exceeds that of the perceptron only by a small margin — up to 3% in the case of Norwegian — while in some instances requiring nearly twice the execution time (e.g., for Italian). Therefore, the perceptron might still be a better choice in terms of efficiency. To better understand why both models show continued improvement over time exclusively for Norwegian, a detailed epoch-by-epoch and tag-specific performance analysis was conducted on the MLR model. This

Table 1: Swedish vs. Italian
(H_0 : MLR is not better than Perceptron)

Parameter	Swedish	Italian
Perceptron Epochs	1	1
MLR Epochs	1	1
MLR Learning Rate	0.2	0.2
samples	10000	10000
n (sample size)	26340	101915
p-value	0.0000	0.0000

Table 2: Norwegian vs. N. Sami
(H_0 : MLR is not better than Perceptron)

Parameter	Norwegian	N. Sami
Perceptron Epochs	10	1
MLR Epochs	10	1
MLR Learning Rate	0.2	0.2
samples	10000	10000
n (sample size)	11681	12958
p-value	0.0000	0.0000

model was selected over the perceptron due to its more stable and interpretable learning trajectory. The analysis revealed that while performance declined for a few morphological tags, it improved for the majority. Notable gains were observed for the following tags: N;MASC;INDF;PL (from 690 to 752 correct predictions), N;NEUT;DEF;PL (from 557 to 594), N;FEM+MASC;DEF;PL (from 142 to 172), N;MASC+FEM;DEF;SG (from 7 to 31), N;MASC+NEUT;DEF;PL (from 0 to 11), and N;MASC+NEUT;INDF;PL (from 0 to 8). A plausible explanation for this pattern may lie in the morphological similarities of suffixes within these tag categories — such as -er, -ere, -ene, and -a — which could cause initial confusion during early training stages. This ambiguity may prompt the

model to rely on additional internal structures for disambiguation, thereby requiring more training time to achieve accurate classification. However, this remains a preliminary hypothesis; a comprehensive linguistic analysis would be necessary to confirm its validity. Detailed graphs and supporting data from this analysis are included in the appendix.

References

- Batsuren, Khuyagbaatar et al. (2022). *UniMorph 4.0: Universal Morphology*. arXiv: 2205.03608 [cs.CL]. URL: <https://arxiv.org/abs/2205.03608>.
- Jurafsky, Daniel and James H. Martin (2025). *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models*. 3rd. Online manuscript released January 12, 2025. URL: <https://web.stanford.edu/~jurafsky/slp3/>.

A Performance of models on Norwegian tags, detailed

A.1

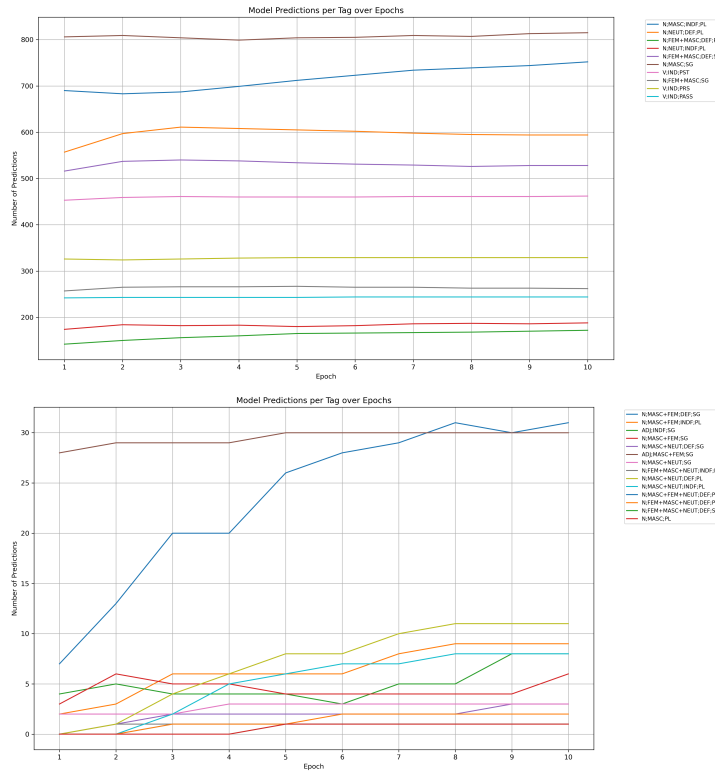


Figure 2: The improvement of MLR performance on Norwegian tags over 10 epochs, divided according to order of magnitude.

A.2

'N:MASC;INDF;PL': [690, 683, 687, 699, 712, 723, 734, 739, 744, 752]	+	62
'N:NEUT;DEF;PL': [557, 597, 611, 608, 605, 602, 598, 595, 594, 594]	+	37
'N:FEM+MASC;DEF;PL': [142,150,156,160,165,166,167,168,170,172]	+	30
'N:MASC+FEM;SG': [7, 13, 20, 20, 26, 28, 29, 31, 30, 31]	+	24
'N:NEUT;INDF;PL': [174, 184, 182, 183, 180, 182, 186, 187, 186, 188]	+	14
'N:FEM+MASC;DEF;SG': [516, 537, 540, 538, 534, 531, 529, 526, 528, 528]	+	12
'N:MASC;SG': [806, 809, 804, 799, 804, 805, 809, 807, 813, 815]	+	9
'V;IND;PST': [453, 459, 461, 460, 460, 460, 461, 461, 461, 462]	+	9
'N:MASC+NEUT;INDF;PL': [2, 3, 6, 6, 6, 6, 6, 8, 9, 9]	+	7
'N:FEM+MASC;SG': [257, 265, 266, 266, 267, 265, 265, 263, 263, 262]	+	5
'ADJ;INDF;SG': [4, 5, 4, 4, 4, 3, 5, 5, 8, 8]	+	4
'V;IND;PRS': [326, 324, 326, 328, 329, 329, 329, 329, 329, 329]	+	3
'N:MASC+FEM;SG': [3, 6, 5, 5, 4, 4, 4, 4, 4, 6]	+	3
'N:MASC+NEUT;DEF;SG': [0, 1, 2, 2, 2, 2, 2, 2, 3, 3]	+	3
'V;IND;PASS': [242, 243, 243, 243, 243, 244, 244, 244, 244, 244]	+	2
'ADJ;MASC+FEM;SG': [28, 29, 29, 29, 30, 30, 30, 30, 30, 30]	+	2
'N:MASC+NEUT;SG': [2, 2, 2, 3, 3, 3, 3, 3, 3, 3]	+	1
N:FEM+MASC+NEUT;INDF;PL: [0, 1, 1, 1, 1, 1, 1, 1, 1, 1]	+	1
N:MASC+NEUT;DEF;PL: [0, 1, 4, 6, 8, 8, 10, 11, 11, 11]	+	11
N:MASC+NEUT;DEF;PL: [0, 0, 2, 5, 6, 7, 7, 8, 8, 8]	+	8
N:MASC+FEM+NEUT;DEF;PL: [0, 0, 1, 1, 1, 1, 1, 1, 1, 1]	+	1
N:FEM+MASC+NEUT;DEF;PL: [0, 0, 1, 1, 1, 2, 2, 2, 2, 2]	+	2
N:FEM+MASC+NEUT;DEF;SG: [0, 0, 0, 0, 1, 1, 1, 1, 1, 1]	+	1
N:MASC;PL: [0, 0, 0, 0, 1, 1, 1, 1, 1, 1]	+	1

252

Figure 3: The improvement of MLR performance on Norwegian tags over 10 epochs shown through the amount of correct predictions.